Project: Analytics on the Growth and Distribution of Electric Vehicles in Washington State.

Introduction:

The rising global demand for sustainable and environmentally friendly transportation has accelerated the transition from conventional internal combustion engine (ICE) vehicles to Electric Vehicles (EVs). This shift is driven by growing concerns about climate change, rising fuel costs, and the need to reduce greenhouse gas (GHG) emissions. Governments worldwide are implementing strong policy incentives, such as tax rebates, emission regulations, and charging infrastructure investments, to encourage EV adoption and support the global move toward cleaner mobility solutions.

Within the United States, Washington State stands out as a pioneer in promoting clean energy and sustainable transportation. With its progressive environmental policies, commitment to carbon neutrality, and strong renewable energy base, Washington has emerged as one of the leading states in EV adoption. The state government has introduced several initiatives, including the Clean Energy Transformation Act (CETA) and the Zero Emission Vehicle (ZEV) mandate, which aim to phase out fossil fuel–based vehicles and encourage the use of electric alternatives. Washington's target to have all new passenger vehicles be zero-emission by 2030 reflects its leadership in advancing transportation electrification.

Problem Statement:

There is limited consolidated analysis on how electric vehicle adoption varies across different counties, years, and manufacturers in Washington State. Understanding these patterns can help in policymaking, infrastructure planning (like charging stations), and future energy strategies.

Electric_Vehicle_Population_Data(1)

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A1 VIN (1-10)

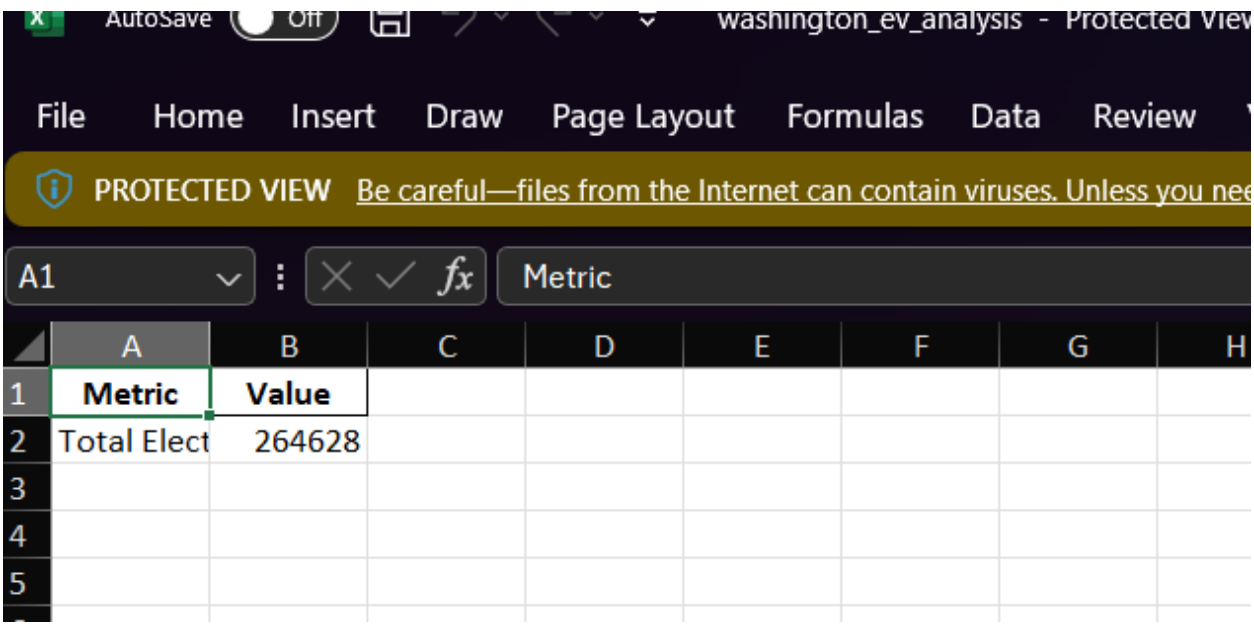
	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V
1	VIN (1-10)	County	City	State	Postal Cod	Model Yea	Make	Model	Electric Ve	Clean Alte	Electric Ra	Base MSRI	Legislative	DOL Vehic	Vehicle Lo	Electric Ut	2020 Census Tract					
2	WA1E2AF	Thurston	Olympia	WA	98512	2024	AUDI	Q5 E	Plug-in Hy	Not eligibl	23	0	22	2.63E+08	POINT (-1	PUGET SO	5.31E+10					
3	WAUUPBF	Yakima	Wapato	WA	98951	2018	AUDI	A3	Plug-in Hy	Not eligibl	16	0	15	3.18E+08	POINT (-1	PACIFICOR	5.31E+10					
4	1N4AZ0CP	King	Seattle	WA	98125	2015	NISSAN	LEAF	Battery Ele	Clean Alte	84	0	46	1.85E+08	POINT (-1	CITY OF SE	5.3E+10					
5	WA1VAAG	King	Kent	WA	98031	2019	AUDI	E-TRON	Battery Ele	Clean Alte	204	0	11	2.59E+08	POINT (-1	PUGET SO	5.3E+10					
6	7SAXCAE5	Snohomisl	Bothell	WA	98021	2022	TESLA	MODEL X	Battery Ele	Eligibility u	0	0	1	2.08E+08	POINT (-1	PUGET SO	5.31E+10					
7	KNDJP3AE	Snohomisl	Lynnwood	WA	98037	2016	KIA	SOUL	Battery Ele	Clean Alte	93	31950	21	2.09E+08	POINT (-1	PUGET SO	5.31E+10					
8	1N4AZ1CP	Snohomisl	Edmonds	WA	98026	2019	NISSAN	LEAF	Battery Ele	Clean Alte	150	0	32	4.78E+08	POINT (-1	PUGET SO	5.31E+10					
9	KNDCC3LC	Snohomisl	Brier	WA	98036	2020	KIA	NIRO	Battery Ele	Clean Alte	239	0	1	2.81E+08	POINT (-1	PUGET SO	5.31E+10					
10	KM8JFDD2	Kitsap	Bremertor	WA	98311	2025	HYUNDAI	TUCSON	Plug-in Hy	Clean Alte	32	0	23	2.81E+08	POINT (-1	PUGET SO	5.3E+10					
11	1C4JJXR6X	Sauk	Spring Gre	WI	53588	2021	JEEP	WRANGLE	Plug-in Hy	Not eligibl	21	0		1.82E+08	POINT (-9	NON WAS	5.51E+10					
12	7SAYGDEE	Snohomisl	Monroe	WA	98272	2023	TESLA	MODEL Y	Battery Ele	Eligibility u	0	0	39	2.29E+08	POINT (-1	PUGET SO	5.31E+10					
13	5YJSA1S29	Snohomisl	Bothell	WA	98021	2015	TESLA	MODEL S	Battery Ele	Clean Alte	208	0	1	1.99E+08	POINT (-1	PUGET SO	5.31E+10					
14	5YJ3E1EAC	King	Seattle	WA	98199	2018	TESLA	MODEL 3	Battery Ele	Clean Alte	215	0	36	1.98E+08	POINT (-1	CITY OF SE	5.3E+10					
15	5YJSA1E1X	Kitsap	Silverdale	WA	98383	2016	TESLA	MODEL S	Battery Ele	Clean Alte	210	0	23	2.73E+08	POINT (-1	PUGET SO	5.3E+10					
16	5UX43EUC	Snohomisl	Mukilteo	WA	98275	2024	BMW	X5	Plug-in Hy	Clean Alte	39	0	21	2.5E+08	POINT (-1	PUGET SO	5.31E+10					
17	5YJSA1E23	Yakima	Toppenish	WA	98948	2018	TESLA	MODEL S	Battery Ele	Clean Alte	249	0	15	1.54E+08	POINT (-1	PACIFICOR	5.31E+10					
18	5YJYGDEE	Snohomisl	Snohomisl	WA	98290	2020	TESLA	MODEL Y	Battery Ele	Clean Alte	291	0	44	2.62E+08	POINT (-1	PUGET SO	5.31E+10					
19	1N4AZ0CP	Kitsap	Bainbridge	WA	98110	2014	NISSAN	LEAF	Battery Ele	Clean Alte	84	0	23	2.51E+08	POINT (-1	PUGET SO	5.3E+10					
20	5YJSA1E23	Thurston	Olympia	WA	98513	2016	TESLA	MODEL S	Battery Ele	Clean Alte	210	0	2	2.4E+08	POINT (-1	PUGET SO	5.31E+10					
21	WA1VABG	Kitsap	Bainbridge	WA	98110	2019	AUDI	E-TRON	Battery Ele	Clean Alte	204	0	23	4697920	POINT (-1	PUGET SO	5.3E+10					
22	5YJSA1E4X	Thurston	Lacey	WA	98516	2016	TESLA	MODEL S	Battery Ele	Clean Alte	210	0	22	3.5E+08	POINT (-1	PUGET SO	5.31E+10					
23	1FADP5CU	Thurston	Olympia	WA	98512	2016	FORD	C-MAX	Plug-in Hy	Not eligibl	19	0	35	1.24E+08	POINT (-1	PUGET SO	5.31E+10					
24	1N4AZ1CP	Thurston	Olympia	WA	98501	2019	NISSAN	LEAF	Battery Ele	Clean Alte	150	0	22	3.47E+08	POINT (-1	PUGET SO	5.31E+10					
25	5YJ3E1EB4	Yakima	Zillah	WA	98953	2019	TESLA	MODEL 3	Battery Ele	Clean Alte	220	0	15	2.68E+08	POINT (-1	PACIFICOR	5.31E+10					
26	1N4BZ0CP	Kitsap	Port Orcha	WA	98366	2016	NISSAN	LEAF	Battery Ele	Clean Alte	84	0	26	1.21E+08	POINT (-1	PUGET SO	5.3E+10					
27	5LMYJ8XY	Kitsap	Port Orcha	WA	98366	2020	LINCOLN	AVIATOR	Plug-in Hy	Not eligibl	21	0	26	2.6E+08	POINT (-1	PUGET SO	5.3E+10					
28	5YJ3E1EB2	King	Burien	WA	98146	2018	TESLA	MODEL 3	Battery Ele	Clean Alte	215	0	34	4.74E+08	POINT (-1	CITY OF SE	5.3E+10					
29	5YJ3E1EB6	King	Kent	WA	98042	2018	TESLA	MODEL 3	Battery Ele	Clean Alte	215	0	47	2.38E+08	POINT (-1	PUGET SO	5.3E+10					
30	WMEFJ9B	King	Kirkland	WA	98034	2017	SMART	FORTWO	Battery Ele	Clean Alte	58	0	45	1.34E+08	POINT (-1	PUGET SO	5.3E+10					
31	WA1AAAG	Kitsap	Port Orcha	WA	98367	2021	AUDI	E-TRON	Battery Ele	Clean Alte	222	0	35	1.57E+08	POINT (-1	PUGET SO	5.3E+10					

Objectives

· To analyze the total number of electric vehicles registered in Washington State.

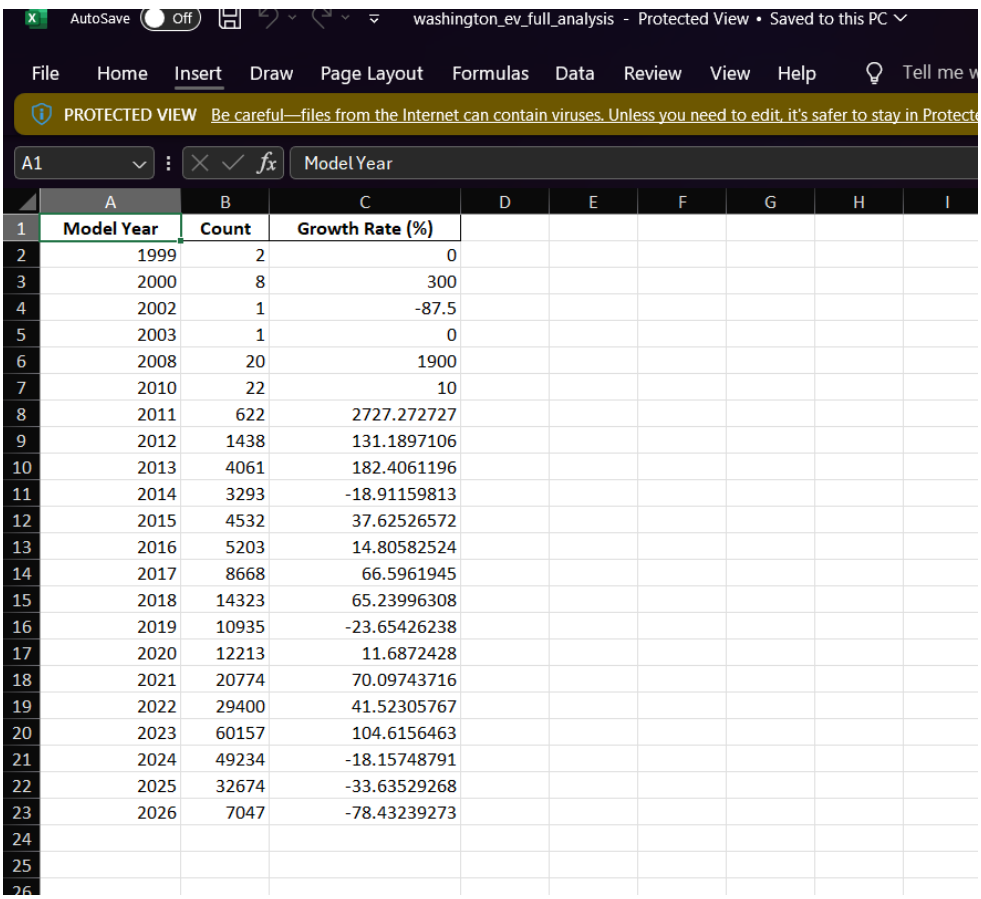
```python
import pandas as pd
import requests
from io import StringIO
# URL for the Washington State Electric Vehicle Population Data (CSV format)
url = "https://data.wa.gov/api/views/f6w7-q2d2/rows.csv?accessType=DOWNLOAD"
# Fetch the data
response = requests.get(url)
data = StringIO(response.text)
# Load the data into a pandas DataFrame
df = pd.read_csv(data)
# Analyze the total number of electric vehicles
total_evs = len(df)
# Print the total number to the console
print(f"Total number of electric vehicles registered in Washington State: {total_evs}")
# Create a simple Excel sheet with the analysis
# We'll create a new DataFrame for the summary
summary_df = pd.DataFrame({
    'Metric': ['Total Electric Vehicles'],
    'Value': [total_evs]
})
# Export to Excel
summary_df.to_excel('washington_ev_analysis.xlsx', index=False)
print("Analysis exported to 'washington_ev_analysis.xlsx'")
```

```
Total number of electric vehicles registered in Washington State: 264628
Analysis exported to 'washington_ev_analysis.xlsx'
```



· To study the year-wise growth trends of EV adoption.

```python
import pandas as pd
import requests
from io import StringIO

# URL for the Washington State Electric Vehicle Population Data (CSV format)
url = "https://data.wa.gov/api/views/f6w7-q2d2/rows.csv?accessType=DOWNLOAD"

# Fetch the data
response = requests.get(url)
data = StringIO(response.text)

# Load the data into a pandas DataFrame
df = pd.read_csv(data)

# Assuming 'Model Year' is the column for the year (adjust if column name differs)
# Group by 'Model Year' and count the number of EVs
yearly_counts = df.groupby('Model Year').size().reset_index(name='Count')

# Sort by year
yearly_counts = yearly_counts.sort_values('Model Year')

# Calculate year-over-year growth rate
yearly_counts['Growth Rate (%)'] = yearly_counts['Count'].pct_change() * 100

# Fill NaN for the first year (no previous year to compare)
yearly_counts['Growth Rate (%)'] = yearly_counts['Growth Rate (%)'].fillna(0)

# Print the year-wise growth trends to the console
print("Year-wise EV Adoption Trends in Washington State:")
print(yearly_counts.to_string(index=False))

# Export to Excel for further analysis
yearly_counts.to_excel('washington_ev_yearly_trends.xlsx', index=False)
print("\nTrends exported to 'washington_ev_yearly_trends.xlsx'")
```



Model Year	Count	Growth Rate (%)
1999	2	0
2000	8	300
2002	1	-87.5
2003	1	0
2008	20	1900
2010	22	10
2011	622	2727.272727
2012	1438	131.1897106
2013	4061	182.4061196
2014	3293	-18.91159813
2015	4532	37.62526572
2016	5203	14.80582524
2017	8668	66.5961945
2018	14323	65.23996308
2019	10935	-23.65426238
2020	12213	11.6872428
2021	20774	70.09743716
2022	29400	41.52305767
2023	60157	104.6156463
2024	49234	-18.15748791
2025	32674	-33.63529268
2026	7047	-78.43239273

· To examine the county-wise and city-wise distribution.

```python
import pandas as pd
import requests
from io import StringIO

# URL for the Washington State Electric Vehicle Population Data (CSV format)
url = "https://data.wa.gov/api/views/f6w7-q2d2/rows.csv?accessType=DOWNLOAD"

# Fetch the data again
response = requests.get(url)
data = StringIO(response.text)

# Load the data into a pandas DataFrame
df =  pd.read_csv("/content/Electric_Vehicle_Population_Data(1)(in).csv")

# Total number of EVs for percentage calculations
total_evs = len(df)

# County-wise distribution
county_distribution = df.groupby('County').size().reset_index(name='EV Count')
county_distribution['Percentage (%)'] = (county_distribution['EV Count'] / total_evs) * 100
county_distribution = county_distribution.sort_values('EV Count', ascending=False)

# City-wise distribution
city_distribution = df.groupby('City').size().reset_index(name='EV Count')
city_distribution['Percentage (%)'] = (city_distribution['EV Count'] / total_evs) * 100
city_distribution = city_distribution.sort_values('EV Count', ascending=False)

# Print top 10 county-wise distribution to console
print("Top 10 County-wise EV Distribution in Washington State:")
print(county_distribution.head(10).to_string(index=False))

# Print top 10 city-wise distribution to console
print("\nTop 10 City-wise EV Distribution in Washington State:")
print(city_distribution.head(10).to_string(index=False))

# Export full distributions to Excel
with pd.ExcelWriter('washington_ev_distribution.xlsx') as writer:
    county_distribution.to_excel(writer, sheet_name='County Distribution', index=False)
    city_distribution.to_excel(writer, sheet_name='City Distribution', index=False)

print("\nFull distributions exported to 'washington_ev_distribution.xlsx'")
```

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A1 | County

	County	EV Count	Percentage (%)
2	King	131179	49.57109603
3	Snohomish	32715	12.36263736
4	Pierce	21855	8.258763245
5	Clark	16232	6.133893617
6	Thurston	9616	3.63378025
7	Kitsap	8942	3.37908309
8	Spokane	7433	2.808848648
9	Whatcom	6488	2.45174358
10	Benton	3653	1.380428375
11	Skagit	3117	1.177879892
12	Island	2918	1.102679989
13	Yakima	1791	0.676799129
14	Chelan	1654	0.625028342
15	Clallam	1610	0.608401227
16	Jefferson	1394	0.526777212
17	Cowlitz	1388	0.524509878
18	Mason	1310	0.495034539
19	San Juan	1236	0.467070756
20	Lewis	1186	0.448176308
21	Franklin	1095	0.413788412
22	Grant	1014	0.383179407
23	Grays Harbor	1007	0.380534184
24	Kittitas	982	0.37108696
25	Walla Walla	718	0.271324274
26	Douglas	587	0.22182082
27	Whitman	536	0.202548483
28	Klickitat	461	0.174206811
29	Okanogan	391	0.147754584
30	Pacific	334	0.126214913
31	Stevens	327	0.12356969

· To compare the popularity of EV brands and models.

```python
import pandas as pd
import requests
from io import StringIO

# URL for the Washington State Electric Vehicle Population Data (CSV format)
url = "https://data.wa.gov/api/views/f6w7-q2d2/rows.csv?accessType=DOWNLOAD"

# Fetch the data again
response = requests.get(url)
data = StringIO(response.text)

# Load the data into a pandas DataFrame
df =  pd.read_csv("/content/Electric_Vehicle_Population_Data(1)(in).csv")
# Total number of EVs for percentage calculations
total_evs = len(df)
# Brand-wise popularity (group by 'Make')
brand_distribution = df.groupby('Make').size().reset_index(name='EV Count')
brand_distribution['Percentage (%)'] = (brand_distribution['EV Count'] / total_evs) * 100
brand_distribution = brand_distribution.sort_values('EV Count', ascending=False)
# Model-wise popularity (group by 'Make' and 'Model')
model_distribution = df.groupby(['Make', 'Model']).size().reset_index(name='EV Count')
model_distribution['Percentage (%)'] = (model_distribution['EV Count'] / total_evs) * 100
model_distribution = model_distribution.sort_values('EV Count', ascending=False)
# Print top 10 brands to console
print("Top 10 EV Brands by Popularity in Washington State:")
print(brand_distribution.head(10).to_string(index=False))
# Print top 10 models to console
print("\nTop 10 EV Models by Popularity in Washington State:")
print(model_distribution.head(10).to_string(index=False))
# Export full distributions to Excel
with pd.ExcelWriter('washington_ev_brand_model_popularity.xlsx') as writer:
    brand_distribution.to_excel(writer, sheet_name='Brand Popularity', index=False)
    model_distribution.to_excel(writer, sheet_name='Model Popularity', index=False)
print("\nFull popularity data exported to 'washington_ev_brand_model_popularity.xlsx'")
```

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A1 | Make

	A	B	C	D	E	F
1	Make	EV Count	rcentage (%)			
2	TESLA	108633	41.05121			
3	CHEVROLE	19051	7.199163			
4	NISSAN	16177	6.11311			
5	FORD	14347	5.421573			
6	KIA	13183	4.98171			
7	TOYOTA	11022	4.165092			
8	BMW	10882	4.112188			
9	HYUNDAI	9278	3.506054			
10	RIVIAN	8118	3.067703			
11	VOLKSWA	7074	2.673187			
12	VOLVO	6899	2.607056			
13	JEEP	6811	2.573802			
14	AUDI	5524	2.087459			
15	CHRYSLER	3874	1.463942			
16	MERCEDE	3081	1.164276			
17	HONDA	2825	1.067536			
18	SUBARU	2588	0.977977			
19	PORSCHE	1868	0.705897			
20	CADILLAC	1828	0.690781			
21	POLESTAR	1511	0.57099			
22	MAZDA	1466	0.553985			
23	LEXUS	1243	0.469716			
24	MINI	1228	0.464048			
25	MITSUBISH	1227	0.46367			
26	FIAT	856	0.323473			
27	DODGE	792	0.299288			
28	LUCID	536	0.202548			
29	GMC	489	0.184788			
30	GENESIS	453	0.171184			
31	LINCOLN	426	0.160981			

· To analyze electric range trends by model year.

```python
import pandas as pd
import requests
from io import StringIO

# URL for the Washington State Electric Vehicle Population Data (CSV format)
url = "https://data.wa.gov/api/views/f6w7-q2d2/rows.csv?accessType=DOWNLOAD"

# Fetch the data
response = requests.get(url)
data = StringIO(response.text)

# Load the data into a pandas DataFrame
df = pd.read_csv(data)

# Filter out rows where Electric Range is 0 or NaN (assuming 0 means not applicable, like for PHEVs)
df = df[df['Electric Range'] > 0]

# Group by 'Model Year' and calculate average electric range
range_trends = df.groupby('Model Year')['Electric Range'].agg(['mean', 'median', 'min', 'max', 'count']).reset_index()
range_trends = range_trends.rename(columns={
    'mean': 'Average Range (miles)',
    'median': 'Median Range (miles)',
    'min': 'Min Range (miles)',
    'max': 'Max Range (miles)',
    'count': 'Number of EVs'
})

# Sort by Model Year
range_trends = range_trends.sort_values('Model Year')

# Calculate year-over-year change in average range
range_trends['YoY Change in Avg Range (%)'] = range_trends['Average Range (miles)'].pct_change() * 100
range_trends['YoY Change in Avg Range (%)'] = range_trends['YoY Change in Avg Range (%)'].fillna(0)

# Print the electric range trends to the console
print("Electric Range Trends by Model Year in Washington State:")
print(range_trends.to_string(index=False))

# Export to Excel for further analysis
range_trends.to_excel('washington_ev_range_trends.xlsx', index=False)
```

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A1 | Model Year

Model Year	Average Range (miles)	Median Range (miles)	Min Range (miles)	Max Range (miles)	Number of EVs	YoY Change in Avg Range (%)
1999	74	74	74	74	2	0
2000	58	58	58	58	8	-21.62162162
2002	95	95	95	95	1	63.79310345
2003	95	95	95	95	1	0
2008	220	220	220	220	19	131.5789474
2010	231.8181818	245	100	245	22	5.371900826
2011	71.33118971	73	35	245	622	-69.22968287
2012	59.80528512	35	6	265	1438	-16.15829575
2013	78.91701551	75	6	208	4061	31.95659106
2014	79.03613726	72	6	208	3293	0.150945581
2015	95.69373345	84	6	208	4532	21.07592396
2016	100.9240823	84	10	210	5203	5.465717159
2017	118.0032303	97	8	238	8668	16.92276772
2018	157.3435274	215	8	249	14322	33.33832225
2019	176.4947397	220	10	289	10931	12.17159206
2020	238.8654417	266	12	337	12136	35.33856138
2021	50.71505269	29	14	233	5029	-78.76835915
2022	28.90588962	30	14	42	4856	-43.00333316
2023	30.50594522	32	13	51	7485	5.535396501
2024	34.36777861	32	6	308	10732	12.65928118
2025	34.90396891	34	1	60	7206	1.560154105
2026	37.49615385	38	24	54	260	7.426619413

· To derive insights that support future planning and policy decisions.

```python
import pandas as pd
import requests
from io import StringIO

# URL for the Washington State Electric Vehicle Population Data (CSV format)
url = "https://data.wa.gov/api/views/f6w7-q2d2/rows.csv?accessType=DOWNLOAD"

# Fetch the data
response = requests.get(url)
data = StringIO(response.text)

# Load the data into a pandas DataFrame
df = pd.read_csv(data)

# Total EVs
total_evs = len(df)

# Year-wise growth (using Model Year as proxy)
yearly_counts = df.groupby('Model Year').size().reset_index(name='Count')
yearly_counts = yearly_counts.sort_values('Model Year')
yearly_counts['Growth Rate (%)'] = yearly_counts['Count'].pct_change() * 100
yearly_counts['Growth Rate (%)'] = yearly_counts['Growth Rate (%)'].fillna(0)
latest_growth = yearly_counts['Growth Rate (%)'].iloc[-1] if len(yearly_counts) > 1 else 0

# County-wise distribution (top 5)
county_dist = df.groupby('County').size().reset_index(name='EV Count')
county_dist['Percentage (%)'] = (county_dist['EV Count'] / total_evs) * 100
county_dist = county_dist.sort_values('EV Count', ascending=False)
top_counties = county_dist.head(5)

# Brand popularity (top 5)
brand_dist = df.groupby('Make').size().reset_index(name='EV Count')
brand_dist['Percentage (%)'] = (brand_dist['EV Count'] / total_evs) * 100
brand_dist = brand_dist.sort_values('EV Count', ascending=False)
top_brands = brand_dist.head(5)

# Electric range trends (average by Model Year, filtered for >0 range)
df_range = df[df['Electric Range'] > 0]
range_trends = df_range.groupby('Model Year')['Electric Range'].mean().reset_index(name='Avg Range (miles)')
```

```python
# Derive insights
insights = [
    f"Total EVs registered: {total_evs}. This indicates the current scale of EV adoption.",
    f"Latest year-over-year growth rate: {latest_growth:.2f}%. Sustained growth suggests increasing demand; plan for infrastructure expansion.",
    f"Top counties by EV count: {', '.join(top_counties['County'].tolist())}. Focus policy incentives here to address regional disparities.",
    f"Top brands: {', '.join(top_brands['Make'].tolist())}. Support popular brands with subsidies or partnerships to accelerate adoption.",
    f"Average annual improvement in electric range: {range_improvement:.2f}%. Increasing range reduces range anxiety; invest in fast-charging networks.",
    "Overall: EV adoption is growing rapidly. Policies should prioritize charging infrastructure in high-adoption areas, incentives for diverse brands, and education on range ben
]

# Print insights to console
print("Key Insights for EV Planning and Policy in Washington State:")
for i, insight in enumerate(insights, 1):
    print(f"{i}. {insight}")

# Create a summary DataFrame for export
summary_df = pd.DataFrame({
    'Insight Category': [
        'Total EVs',
        'Growth Rate',
        'Top Counties',
        'Top Brands',
        'Range Improvement',
        'Overall Recommendation'
    ],
    'Details': [
        f"{total_evs} registered EVs",
        f"Latest YoY: {latest_growth:.2f}%",
        f"{', '.join(top_counties['County'].tolist())} ({', '.join([f'{p:.1f}%' for p in top_counties['Percentage (%)']])})",
        f"{', '.join(top_brands['Make'].tolist())} ({', '.join([f'{p:.1f}%' for p in top_brands['Percentage (%)']])})",
        f"Avg annual: {range_improvement:.2f}%",
        "Prioritize infrastructure, incentives, and education."
    ]
})

# Export to Excel
summary_df.to_excel('washington_ev_policy_insights.xlsx', index=False)

print("\nInsights summary exported to 'washington_ev_policy_insights.xlsx'")
```

PROTECTED VIEW Be careful—files from the Internet can contain viruses. Unless you need to edit, it's safer to stay in Protecte

A1 | Insight Category

	A	B	C	D	E	F	G	H
1	**Insight Category**	**Details**						
2	Total EVs	264628 registered EVs						
3	Growth Rate	Latest YoY: -78.43%						
4	Top Counties	King, Snohomish, Pierce, Clark, Thurston (49.6%, 12.4%, 8.3%, 6.1%, 3.6%)						
5	Top Brands	TESLA, CHEVROLET, NISSAN, FORD, KIA (41.1%, 7.2%, 6.1%, 5.4%, 5.0%)						
6	Range Improvement	Avg annual: 7.41%						
7	Overall Recommendation	Prioritize infrastructure, incentives, and education.						
8								
9								
10								